

AI-Based Onion Quality Inspection & Automated Grading Report  
SIH Technical Produce Evaluation • Team: THE DEBUGGERS

1. Executive Batch Inspection Summary

| TOTAL ONIONS | QUALITY SCORE | HEALTHY BULBS | DEFECT RATE | AVG SIZE     |
|--------------|---------------|---------------|-------------|--------------|
| 1            | 60.0 / 100    | 2 / 1         | 0.0%        | Uncalibrated |

2. Prototype Grade Distribution Breakdown

| Grade Tier           | Count | Share  | Commercial Standard Description                                                           |
|----------------------|-------|--------|-------------------------------------------------------------------------------------------|
| Grade A (Premium)    | 0     | 0.0%   | Sound health, uniform size (45–75 mm when calibrated), negligible surface defect (≤5%).   |
| Grade B (Commercial) | 2     | 200.0% | Good commercial quality, minor size variance (35–45 or 75–90 mm), minor blemishes (≤10%). |
| Grade C (Marginal)   | 0     | 0.0%   | Marginal quality, undersized (<35 mm), oversized (>90 mm), or moderate blemishes (≤20%).  |
| Reject (Unsound)     | 0     | 0.0%   | Severe visible rot, extensive surface decay (>20%), or major pathological defects.        |

3. Metrology & Physical Scale Calibration

|                      |               |                      |                    |
|----------------------|---------------|----------------------|--------------------|
| Calibration State:   | UNCALIBRATED  | Ref Marker Detected: | NO (Marker absent) |
| Known Reference Std: | None provided | Scale Factor:        | N/A (N/A)          |

PIXEL MEASUREMENT ONLY: Physical measurements unavailable — reference-object calibration required. No reference standard disc was verified; millimeter dimensions are omitted to prevent metric fabrication.

5. Individual Onion Detection & Morphometry Records

Detailed inspection record for each localized onion instance, including neural classification confidence, morphological parameters, and deterministic grading rationale.

|               |                                |                                                                                                                                                      |                               |                                      |
|---------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------|
| No Crop<br>#1 | <b>Bulb #1 — Yellow-Onion</b>  | <b>Grade:</b> <b>GRADE B</b>                                                                                                                         | <b>Health:</b> <b>Healthy</b> | <b>Review:</b><br>REVIEW_RECOMMENDED |
|               | <b>Seg Conf:</b> 91.0%         | <b>Health Conf:</b> 91.0%                                                                                                                            | <b>Combined Conf:</b> 91.0%   | <b>Defect Area:</b> 0.0%             |
|               | <b>PIXEL MEASUREMENT:</b>      | <b>Eq. Diameter:</b> 110.0 px   <b>Area:</b> 9500 px²   <b>Perimeter:</b> 380.0 px                                                                   |                               |                                      |
|               | <b>CALIBRATED MEASUREMENT:</b> | <b>Physical measurements unavailable — reference-object calibration required.</b>                                                                    |                               |                                      |
|               | <b>Grading Rationale:</b>      | <ul style="list-style-type: none"><li>• Healthy bulb</li><li>• Physical size unavailable because calibration reference was not configured.</li></ul> |                               |                                      |

|               |                                |                                                                                                                                                      |                               |                                      |
|---------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------|
| No Crop<br>#1 | <b>Bulb #1 — Yellow-Onion</b>  | <b>Grade:</b> <b>GRADE B</b>                                                                                                                         | <b>Health:</b> <b>Healthy</b> | <b>Review:</b><br>REVIEW_RECOMMENDED |
|               | <b>Seg Conf:</b> 91.0%         | <b>Health Conf:</b> 91.0%                                                                                                                            | <b>Combined Conf:</b> 91.0%   | <b>Defect Area:</b> 0.0%             |
|               | <b>PIXEL MEASUREMENT:</b>      | <b>Eq. Diameter:</b> 110.0 px   <b>Area:</b> 9500 px²   <b>Perimeter:</b> 380.0 px                                                                   |                               |                                      |
|               | <b>CALIBRATED MEASUREMENT:</b> | <b>Physical measurements unavailable — reference-object calibration required.</b>                                                                    |                               |                                      |
|               | <b>Grading Rationale:</b>      | <ul style="list-style-type: none"><li>• Healthy bulb</li><li>• Physical size unavailable because calibration reference was not configured.</li></ul> |                               |                                      |

6. Technical Limitations & Engineering Disclaimers

- 1. Visible Surface Inspection:** RGB optical inspection captures only visible external outer tunic and epidermal features. Subsurface physiological decay, internal core rot, black mold beneath dry tunics, and bacterial soft rot cannot be evaluated without destructive cutting or non-visible spectral scanning.

**2. Internal Rot Limitation:** A 'Healthy' external surface classification does not guarantee the absence of interior fungal or bacterial degradation.

**3. Scale Calibration Requirement:** Absolute millimeter measurements require an explicit planar reference standard. When uncalibrated, only raw pixel morphometry is recorded; metric physical dimensions are never fabricated.

**4. Prototype Grading Notice:** Automated grading tiers (Grade A/B/C/Reject) are derived from an engineering prototype rule engine for Smart India Hackathon technical evaluation and decision support. They do not constitute statutory agricultural certification (such as AGMARK, NAFED, or export grade certificates).

**5. Confidence Metric:** Model confidence denotes neural network softmax certainty on visible visual patterns, not verified laboratory accuracy or shelf-life durability.

**6. Decision Support:** Results are intended for assisted inspection and preliminary screening. Critical commercial batches require certified laboratory metrology and expert human inspection.